

THREE COLOURS FOR THE PLANE OVER A NUMBER FIELD

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ABSTRACT. For a field F of characteristic 0 let $\chi(F^2)$ be the chromatic number of the graph on F^2 that joins x and y when $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$. We prove that for every number field F , $\chi(F^2) \leq 3$ if and only if some prime of F above 2 ramifies in $F(i)$ or some prime of F above 3 has residue degree 1. The “if” part is classical (colourings through a residue field); the “only if” part extends to all number fields our earlier result that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for $d \equiv 11 \pmod{12}$. It answers, for three colours, the question whether the chromatic number of the plane over a number field is decided by a single completion, and it classifies the finite extensions K of $\mathbb{Q}_p$: $\chi(K^2)$ is 2, 3 or at least 4 according to explicit local conditions. We also give an elementary family (every real field containing $\sqrt{a}$ and $\sqrt{b}$ with $a \equiv 2 \pmod{3}$ and $b \equiv 7 \pmod{8}$ needs four colours), new examples such as $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{7})^2) = 4$, and, as far as we know, the first non-integral value of the circular chromatic number of the plane over a number field: $\chi_c(\mathbb{Q}(\sqrt{11})^2) = \chi_c(\mathbb{Q}(\sqrt{35})^2) = 7/2$ (and $\chi_c(\mathbb{Q}_7^2) = 7/2$), although these planes have chromatic number 4. Locally constant colourings of one completion give no upper bound on χ_c below 4 other than 2, 3 and $7/2$. Finally, $\chi_c(F^2) \geq 56/17$ whenever $\chi(F^2) \geq 4$, so the circular chromatic number of the plane over a number field never lies in $(3, 56/17)$; with a computer-assisted step, not even in $(3, 3.3315]$.

1. INTRODUCTION

Let F be a field of characteristic 0. The *plane over F* is the graph on F^2 in which x and y are adjacent when $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$, and $\chi(F^2)$ is its chromatic number. For $F = \mathbb{R}$ this is the Hadwiger–Nelson problem, and $5 \leq \chi(\mathbb{R}^2) \leq 7$ [5]. Over number fields the question goes back to Woodall ($\chi(\mathbb{Q}^2) = 2$ [22]) and was studied by Johnson [15], Fischer [7], Benda and Perles [1], Moorhouse [19] and Madore [17]. All upper bounds below 7 known to us come from a colouring that is locally constant at one completion F_v : reduction modulo a power of a prime. Writing $L = F(i)$, the unit vectors of F^2 are the elements of norm one of L , and at a place where $x^2 + y^2$ is anisotropic they reduce into a finite group of roots of unity of the residue field. Two such reductions give colourings with at most three colours:

- (a) at a prime of F above 2 that ramifies in $F(i)$, every unit vector is $\equiv 1$ modulo the maximal ideal, which gives a 2-colouring (Johnson; Fischer [7, Thm. 8]; Moorhouse [19]; [18, Theorem A’]);
- (b) at a prime of F above 3 of residue degree 1, the unit vectors reduce into $\mu_4 \subset \mathbb{F}_9$, and $a + bi \mapsto a + b$ gives a 3-colouring (Madore [17, Cor. 3.4]; for real quadratic fields Moorhouse [19, Lemma 8.2 and Cor. 8.3] and Fischer [7, Thm. 9]).

In [13] we proved that (a) is also necessary for $\chi(F^2) = 2$. Here we prove the same at three colours.

Theorem B. *Let F be a number field. Then $\chi(F^2) \leq 3$ if and only if (a) or (b) holds; and $\chi(F^2) \leq 2$ if and only if (a) holds.*

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These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the proofs, wrote them and the programs that check them, and helped to draft this note. Separate AI agents refereed the proofs, with their own programs. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

For real quadratic fields $\mathbb{Q}(\sqrt{d})$, d squarefree, (a) holds iff $d \not\equiv 3 \pmod{4}$ and (b) iff $d \not\equiv 2 \pmod{3}$, so Theorem B contains the main result of [11] (Theorem 1 and Corollary 2 there): $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ iff $d \equiv 11 \pmod{12}$. Whether (a) or (b) holds depends only on $F \otimes \mathbb{Q}_2$ and $F \otimes \mathbb{Q}_3$, and is computed from a defining polynomial (the relative discriminant of $F(i)/F$, and the residue degrees above 3). By the theorem of de Bruijn and Erdős [4], $\chi(F^2) \geq 4$ if and only if some finite unit-distance graph in F^2 needs four colours, and Theorem B says that this happens unless one of the completions of F has a locally constant 3-colouring: a local-global principle for three colours. The same proof decides the local fields (Corollary 3): for a finite extension K of $\mathbb{Q}_p$, $\chi(K^2) = 2$ if $p = 2$ and $K(i)/K$ is ramified, $\chi(K^2) = 3$ if $p = 3$ and K has residue degree 1, and $\chi(K^2) \geq 4$ otherwise; for instance the plane over the unramified cubic extension of $\mathbb{Q}_3$ needs at least four colours, although $x^2 + y^2$ is anisotropic there, as over $\mathbb{Q}_3$.

Two families of fields were known to need four colours: those containing $\sqrt{d}$ with $d \equiv 11 \pmod{12}$ [11], and those containing $\sqrt{3}$ and $\sqrt{q}$ with $q \equiv 2 \pmod{3}$, by chains of unit rhombi ([9], `notes/local_colourings.md`, Section 11, Theorem 5). Theorem B adds, for example, $\mathbb{Q}(\sqrt{2}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{31})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{55})$, which contain no unit triangle and whose quadratic subfields all have $\chi \leq 3$, and $\mathbb{Q}(2 \cos \frac{2\pi}{7}, \sqrt{7})$, of degree 6, where 3 has residue degree 3. For many fields an explicit proof is short (Proposition 2), and for $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ we give a certificate checked by computer that does not depend on the general argument.

Theorem C. $\chi_c(\mathbb{Q}(\sqrt{11})^2) = \chi_c(\mathbb{Q}(\sqrt{35})^2) = \chi_c(\mathbb{Q}_7^2) = 7/2$, while the three chromatic numbers are 4. Moreover no plane over a number field has circular chromatic number strictly between 2 and 3.

Theorem D. Let F be a number field with $\chi(F^2) \geq 4$. Then $\chi_c(F^2) \geq 56/17 \approx 3.294$. With a computer-assisted step (exact computations, made by three programs written separately), $\chi_c(F^2) \geq 10^7/3001611 > 3.3315$.

So the circular chromatic number of the plane over a number field is 2, 3 or at least $56/17$ (larger than 3.3315, with the computer-assisted step); we do not know whether a value in $(3, 7/2)$ occurs (Section 11).

The tools are those of [11]: a characterisation of 3-colourings of Cayley graphs of abelian groups by characters (Theorem W^+ , Section 2), and the exact shape of the characters along the rational rotations whose denominators are powers of 5. These rotations act as a probe: along them, a character that keeps every unit vector in $[1/3, 2/3]$ is, up to a bounded error, either a 2-adic “half” character or a 3-adic character of level one, and a compactness argument turns this into the dichotomy between the places above 2 and those above 3. For Theorem D we show that the probe keeps this shape, with a small error also for type q , when $[1/3, 2/3]$ is replaced by $[\theta, 1 - \theta]$ with $\theta > 17/56$ (Proposition 7).

Sections 2–5 prove Theorem B; Section 6 draws the consequences (circular colourings below 3, fields split above 2 and 3, local fields); Section 7 gives the explicit family; Section 8 proves Theorem C and shows that the restrictions to F^2 of locally constant colourings of a completion F_v^2 give no upper bound on $\chi_c(F^2)$ below 4 other than 2, 3 and $7/2$ (Corollary 5); Section 9 proves Theorem D; Section 10 lists the computer checks, and Section 11 asks whether 2, 3 and $7/2$ are the only values below 4 for number fields. The programs and certificates are in [9].

2. PRELIMINARIES

Characters and colourings. For an abelian group Γ and $S = -S \subseteq \Gamma$, $\text{Cay}(\Gamma, S)$ is the graph on Γ in which $x \sim y$ when $y - x \in S$. A *character* of Γ is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$, and $\|t\|$ is the distance from t to the nearest integer. For integers $p \geq 2q \geq 1$, the circular clique $K_{p/q}$ has vertex set $\mathbb{Z}/p$, with $a \sim b$ when $q \leq (b - a \bmod p) \leq p - q$, and the circular chromatic number $\chi_c(G)$ is the infimum of the p/q such that G has a homomorphism to $K_{p/q}$.

Theorem W^+ ([14, Theorem 1]). Let $p \geq 2q \geq 2$ with $p < 4q$. Then $\text{Cay}(\Gamma, S)$ has a homomorphism to $K_{p/q}$ if and only if some character ξ of Γ satisfies $\xi(s) \in [q/p, 1 - q/p] + \mathbb{Z}$ for every $s \in S$. In particular $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps S into $[1/3, 2/3] + \mathbb{Z}$.

The “if” direction is classical (colour x by $[p\xi(x)]$); the “only if” direction is the content of [14], where it is also proved in Lean for every abelian group Γ and finite S . We apply it to the groups $\Gamma = \mathbb{Z}U$ spanned by finite sets U of unit vectors, and in Section 8 to finite groups $O_w/\mathfrak{m}^k$.

The plane as a Cayley graph. Let F be a number field with $i \notin F$, and $L = F(i)$. The map $(x, y) \mapsto x + iy$ identifies F^2 with L , and $x^2 + y^2 = z\bar{z}$ for $z = x + iy$, where $z \mapsto \bar{z}$ is the non-trivial automorphism of L/F . So the unit vectors form the group $T = \{z \in L : z\bar{z} = 1\}$, and the plane over F is $\text{Cay}(L, T)$. As a $\mathbb{Q}(i)$ -vector space L has dimension $n = [F : \mathbb{Q}]$, and $\text{Tr} = \text{Tr}_{L/\mathbb{Q}(i)}$ gives a non-degenerate pairing $L \times L \rightarrow \mathbb{Q}(i)$.

The probe. Let $N = 5^k$, $k \geq 1$, let G_N be the set of $\gamma \in \mathbb{Q}(i)$ with $\gamma\bar{\gamma} = 1$ and $N\gamma \in \mathbb{Z}[i]$ (the rational rotations whose denominator divides N), $I = [1/3, 2/3]$, and

$$S_N = \{c \in \mathbb{C} : \text{Re}(\bar{c}\gamma) \in I + \mathbb{Z} \text{ for every } \gamma \in G_N\}.$$

Let $E_c = \frac{1+i}{2} + \mathbb{Z}[i]$, $E_q = \{\frac{\alpha+\beta i}{3} : \alpha, \beta \in \{1, 2\}\} + \mathbb{Z}[i]$, $E = E_c \cup E_q$, and $r = \sqrt{10}/18$. Elements of E_c have *type c*, those of E_q *type q*.

Proposition 1 ([11, Proposition 9]). *Every $c \in S_N$ can be written $c = N\varepsilon + x$, either with $\varepsilon \in E_c$ and $|x| \leq r$, or with $\varepsilon \in E_q$ and $x = 0$.*

Lemma 1 (exact relations; cf. [11, Lemma 10]). *Let $\sigma_j = N\varepsilon_j + x_j \in S_N$ be as in Proposition 1 ($1 \leq j \leq \ell$), and let $\mu_1, \dots, \mu_\ell \in \mathbb{Z}[i]$ satisfy $\sum_j \mu_j \sigma_j = 0$ and $N > 6r \sum_j |\mu_j|$. Then $\sum_j \mu_j \varepsilon_j = 0$.*

Proof. $\sum_j \mu_j N\varepsilon_j = -\sum_j \mu_j x_j$. The left side lies in $\frac{N}{6}\mathbb{Z}[i]$, because $6E \subseteq \mathbb{Z}[i]$, and the nonzero elements of $\frac{N}{6}\mathbb{Z}[i]$ have modulus at least $\frac{N}{6}$; the right side has modulus at most $r \sum_j |\mu_j| < \frac{N}{6}$. So both sides vanish. $\square$

Only two properties of E are used below. Let $\mathbb{F}_9 = \mathbb{Z}[i]/3$ and $A' = \{\pm 1 \pm i\} \subset \mathbb{F}_9$, the set of non-squares of $\mathbb{F}_9$ (equivalently, of the $y \in \mathbb{F}_9$ with $y^4 = -1$). Then

$$(1) \quad \varepsilon \in E_c \Rightarrow v_{1+i}(\varepsilon) = -1 \text{ and } v_3(\varepsilon) \geq 0; \quad \varepsilon \in E_q \Rightarrow v_{1+i}(\varepsilon) \geq 0 \text{ and } 3\varepsilon \bmod 3 \in A'.$$

In particular every $\varepsilon \in E$ has $v_{1+i}(\varepsilon) \geq -1$ and $v_3(\varepsilon) \geq -1$, with equality in the second exactly for type q.

3. FROM COLOURINGS TO A LINEAR PROBLEM OVER $\mathbb{Q}(i)$

Lemma 2. *Let F be a number field with $i \notin F$, and suppose that F^2 is 3-colourable. Then for every finite $V \subset T$ there is $\beta \in L$ with $\text{Tr}(\beta v) \in E$ for every $v \in V$.*

Proof. We may enlarge V , so we assume that it contains a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n$ of L . Such a basis exists in T : the $\mathbb{Q}(i)$ -span W of T is a subring of L , as T is a group; being finite-dimensional over $\mathbb{Q}(i)$ and contained in a field, it is a field. For $t \in F$ the element $z = (t+i)/(t-i)$ lies in T , so W contains $(z-1)^{-1}$ and $t = i(z+1)/(z-1)$; hence $W = L$.

For $v \in V$ write $D_v v = \sum_j \mu_{v,j} b_j$ with $D_v \in \mathbb{Z} \setminus \{0\}$ and $\mu_{v,j} \in \mathbb{Z}[i]$, and choose $N = 5^k$ with $N > 6r \max_v (|D_v| + \sum_j |\mu_{v,j}|)$. The set $U = G_N V$ is finite, symmetric ($-1 \in G_N$) and consists of unit vectors, so $\text{Cay}(\mathbb{Z}U, U)$ is a subgraph of $\text{Cay}(L, T)$ and is 3-colourable. By Theorem W⁺ there is a character ξ of $\mathbb{Z}U$ with $\xi(U) \subseteq I + \mathbb{Z}$. Extend ξ to L (the group $\mathbb{R}/\mathbb{Z}$ is divisible) and restrict it to a lattice $\Lambda = \frac{1}{M} \bigoplus_j \mathbb{Z}[i] f_j \supseteq U$, where $f_1, \dots, f_n$ is a $\mathbb{Q}$ -basis of F , so that $L = \bigoplus_j \mathbb{Q}(i) f_j$. A character of $\Lambda \cong \mathbb{Z}^{2n}$ lifts to a real-linear form, and every real-linear form on $\mathbb{C}^n$ is the real part of a $\mathbb{C}$ -linear one; so there is a $\mathbb{Q}(i)$ -linear map $\varphi : L \rightarrow \mathbb{C}$ with $\xi(w) = \text{Re } \varphi(w) \bmod 1$ for $w \in \Lambda$. For $v \in V$ and $\gamma \in G_N$ we get $\xi(\gamma v) = \text{Re}(\gamma \varphi(v))$, so $\overline{\varphi(v)} \in S_N$, and $\varphi(v) \in S_N$ because G_N is closed under conjugation. Write $\varphi(v) = N\varepsilon_v + x_v$ as in Proposition 1. Applying φ to $D_v v - \sum_j \mu_{v,j} b_j = 0$ and Lemma 1 give $D_v \varepsilon_v = \sum_j \mu_{v,j} \varepsilon_{b_j}$. Let $\beta \in L$ be the element with $\text{Tr}(\beta b_j) = \varepsilon_{b_j}$ for all j . Then $D_v \text{Tr}(\beta v) = \sum_j \mu_{v,j} \varepsilon_{b_j} = D_v \varepsilon_v$, so $\text{Tr}(\beta v) = \varepsilon_v \in E$ for every $v \in V$. $\square$

Lemma 2 is also a practical test: if for some finite $V \subset T$ no such β exists, then F^2 is not 3-colourable. Whether β exists is a finite computation (Section 10).

4. LOCALISATION AT 2 AND 3

For a prime p let $L_p = L \otimes_{\mathbb{Q}} \mathbb{Q}_p = \prod_{w|p} L_w$, and for a place u of F let $L_u = L \otimes_F F_u$ and $T(F_u) = \{z \in L_u : z\bar{z} = 1\}$. Put $T_p = \prod_{u|p} T(F_u)$. Since 2 ramifies and 3 is inert in $\mathbb{Q}(i)$, $\mathbb{Q}(i) \otimes \mathbb{Q}_2 = \mathbb{Q}_2(i)$ and $\mathbb{Q}(i) \otimes \mathbb{Q}_3 = \mathbb{Q}_9$, the unramified quadratic extension of $\mathbb{Q}_3$; the trace extends to $\text{Tr} : L_2 \rightarrow \mathbb{Q}_2(i)$ and $\text{Tr} : L_3 \rightarrow \mathbb{Q}_9$. Let $O_2 = \mathbb{Z}_2[i]$, $O_3 = \mathbb{Z}_9$ (the integers of $\mathbb{Q}_9$), and $\mathcal{A} = \{\frac{\pm 1 \pm i}{3}\} + O_3$. By (1), $E_c \subseteq (1+i)^{-1}O_2^\times \cap O_3$ and $E_q \subseteq O_2 \cap \mathcal{A}$.

Lemma 3 (weak approximation for T). *For every finite set Σ of places of F , T is dense in $\prod_{u \in \Sigma} T(F_u)$.*

Proof. At a place u with $i \notin F_u$, $t \mapsto (t+i)/(t-i)$ is a homeomorphism from F_u onto $T(F_u) \setminus \{1\}$, with inverse $z \mapsto i(z+1)/(z-1)$. At a place with $i \in F_u$, $L_u \cong F_u \times F_u$, $T(F_u) = \{(s, s^{-1})\}$, and $t \mapsto (s, s^{-1})$ with $s = (t+i)/(t-i)$, $i^2 = -1$, is a homeomorphism from $F_u \setminus \{\pm i\}$ onto $T(F_u) \setminus \{1\}$. These maps send F into T , F is dense in $\prod_{u \in \Sigma} F_u$, and $T(F_u)$ has no isolated points. $\square$

Lemma 4. *If F^2 is 3-colourable and $i \notin F$, then one of the following holds:*

- (c) *there is $\beta_2 \in L_2$ with $v_{1+i}(\text{Tr}(\beta_2 z)) = -1$ for every $z \in T_2$;*
- (q) *there is $\beta_3 \in L_3$ with $\text{Tr}(\beta_3 z) \in \mathcal{A}$ for every $z \in T_3$.*

Proof. Fix a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n \subset T$ of L (proof of Lemma 2), and let

$$K_2 = \{\beta \in L_2 : \text{Tr}(\beta b_j) \in (1+i)^{-1}O_2 \ \forall j\}, \quad K_3 = \{\beta \in L_3 : \text{Tr}(\beta b_j) \in \frac{1}{3}O_3 \ \forall j\}.$$

They are compact, since $\beta \mapsto (\text{Tr}(\beta b_j))_j$ is a linear isomorphism onto $\mathbb{Q}_2(i)^n$, resp. $\mathbb{Q}_9^n$. On the compact space $\Omega = \{c, q\}^T \times K_2 \times K_3$ consider, for $v \in T$, the condition $P_v(\tau, \beta_2, \beta_3)$:

$$\tau(v) = c, \ v_{1+i}(\text{Tr}(\beta_2 v)) = -1, \ \text{Tr}(\beta_3 v) \in O_3; \quad \text{or} \quad \tau(v) = q, \ \text{Tr}(\beta_2 v) \in O_2, \ \text{Tr}(\beta_3 v) \in \mathcal{A}.$$

Each P_v defines a clopen subset of Ω . For a finite $V \supseteq \{b_1, \dots, b_n\}$, Lemma 2 gives $\beta \in L$; its images in L_2 and L_3 lie in K_2 and K_3 , and with $\tau(v)$ the type of $\text{Tr}(\beta v)$ the triple satisfies P_v for every $v \in V$, by (1). So the closed sets $\{P_v \text{ for all } v \in V\}$ have the finite intersection property, and some $(\tau, \beta_2, \beta_3) \in \Omega$ satisfies P_v for every $v \in T$.

For $z \in T_2$ let $a(z)$ be the statement $v_{1+i}(\text{Tr}(\beta_2 z)) = -1$ and $a'(z)$ the statement $\text{Tr}(\beta_2 z) \in O_2$; for $y \in T_3$ let $b(y)$ be $\text{Tr}(\beta_3 y) \in O_3$ and $b'(y)$ be $\text{Tr}(\beta_3 y) \in \mathcal{A}$. They are locally constant, a and a' exclude each other, and so do b and b' . The set

$$Z = \{(z, y) \in T_2 \times T_3 : (a(z) \wedge b(y)) \vee (a'(z) \wedge b'(y))\}$$

is clopen and contains the diagonal image of T , which is dense by Lemma 3; so $Z = T_2 \times T_3$. If $a(z_0)$ holds for some z_0 , then $b(y)$ holds for every y , so b' holds nowhere, and $a(z)$ holds for every z : this is (c). Otherwise a' holds everywhere, and then b' holds everywhere: this is (q). $\square$

Case (c). For $z \in L$ let $g(z) = (1+i) \text{Tr}(\beta_2 z) \in \mathbb{Q}_2(i)$. Fix a representative ρ of each class of $\mathbb{Q}_2(i)$ modulo O_2 , and colour z by $(g(z) - \rho) \bmod (1+i) \in O_2/(1+i) = \mathbb{F}_2$, where ρ represents the class of $g(z)$. A unit step $v \in T$ adds $g(v) \in O_2^\times$ to $g(z)$: it keeps the class modulo O_2 and changes the residue modulo $1+i$. This is a proper 2-colouring of $L \cong F^2$, so by [13, Theorem 1] some prime of F above 2 ramifies in $F(i)$.

5. CASE (Q): THE PLACES ABOVE 3

Let u be a place of F above 3, $f = f_u$ its residue degree, $q = 3^f$, and $g_u(z) = \text{Tr}_{L_u/\mathbb{Q}_9}(\beta_u z)$, where $\beta_3 = (\beta_u)_u$; then $\text{Tr}(\beta_3 y) = \sum_u g_u(y_u)$ for $y = (y_u) \in T_3$.

Even f . Then $\mathbb{F}_9$ lies in the residue field of F_u , so $i \in F_u$, $L_u \cong F_u \times F_u$ and $T(F_u) = \{(s, s^{-1}) : s \in F_u^\times\}$. Write $g_u(s, t) = \ell_1(s) + \ell_2(t)$ with $\mathbb{Q}_3$ -linear $\ell_1, \ell_2 : F_u \rightarrow \mathbb{Q}_9$. In case (q), $g_u(y_u)$ stays in a bounded set when y_u varies and the other components are fixed. If $\ell_1(s_0) \neq 0$, then $g_u(3^{-m}s_0, 3^m s_0^{-1}) = 3^{-m}\ell_1(s_0) + 3^m\ell_2(s_0^{-1})$ is unbounded as $m \rightarrow \infty$; so $\ell_1 = 0$, likewise $\ell_2 = 0$, and $g_u = 0$.

Odd f . Then $L_u = F_u(i) = F_u\mathbb{Q}_9$ is a field, unramified of degree 2 over F_u , with residue field $\mathbb{F}_{q^2}$; its non-trivial automorphism σ acts on $\mathbb{Q}_9$ as the Frobenius ($\sigma(i) = -i$). The group $T(F_u)$

lies in $O_{L_u}^\times$, contains $\mu_4 \subset \mathbb{Q}_9$ and the Teichmüller lifts of $\mu_{q+1} \subset \mathbb{F}_{q^2}^\times$ (as $\zeta\sigma(\zeta) = \zeta^{1+q}$), and reduces onto μ_{q+1} . Since g_u is $\mathbb{Q}_9$ -linear, $g_u(\epsilon z) = \epsilon g_u(z)$ for $\epsilon \in \mu_4$.

Lemma 5 (one place). *In case (q) there are a place $u_0 \mid 3$ of odd residue degree and $\beta \in L_{u_0}$ with $\text{Tr}_{L_{u_0}/\mathbb{Q}_9}(\beta z) \in \mathcal{A}$ for every $z \in T(F_{u_0})$.*

Proof. Only the places of odd residue degree contribute. For such a place u and $z, z' \in T(F_u)$, varying y_u alone gives $g_u(z) - g_u(z') \in \mathcal{A} - \mathcal{A} \subseteq \frac{1}{3}O_3$. With $z' = 1$ and $z = i$ this says $(i-1)g_u(1) \in \frac{1}{3}O_3$, and $i-1$ is a unit at 3; so $g_u(T(F_u)) \subseteq \frac{1}{3}O_3$. Let $Z_u \subseteq \mathbb{F}_9$ be the set of residues of $3g_u(z)$, $z \in T(F_u)$; it is stable under μ_4 , and (q) says that $\sum_u z_u \in A'$ for every choice of $z_u \in Z_u$. Suppose two places $u \neq u'$ had nonzero elements $w \in Z_u$ and $w' \in Z_{u'}$. Fix $z_{u''} \in Z_{u''}$ at the other places and put $c = \epsilon' w' + \sum z_{u''}$ with $\epsilon' \in \mu_4$. The four distinct elements $c + \epsilon w$, $\epsilon \in \mu_4$, lie in A' , which has four elements; so they are all of A' , and summing them gives $4c = c = \sum A' = 0$. As this holds for every $\epsilon' \in \mu_4$, $w' = iw'$, so $w' = 0$: a contradiction. Since $0 \notin A'$, exactly one place u_0 has $Z_{u_0} \neq \{0\}$; the other g_u take values in O_3 , and $\beta = \beta_{u_0}$ works, because $\mathcal{A} + O_3 = \mathcal{A}$. $\square$

Lemma 6 (level one). *Let $u \mid 3$ have odd residue degree f , and let $\beta \in L_u$ satisfy $\text{Tr}_{L_u/\mathbb{Q}_9}(\beta z) \in \mathcal{A}$ for every $z \in T(F_u)$. Then $\lambda(x) = 3 \text{Tr}_{L_u/\mathbb{Q}_9}(\beta x) \bmod 3$ is defined on O_{L_u} and vanishes on its maximal ideal $\mathfrak{m}$, and the induced $\mathbb{F}_9$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ satisfies $\lambda_1(\mu_{q+1}) \subseteq A'$.*

Proof. The closed additive span R of $T(F_u)$ is a closed subring of O_{L_u} . It contains $\mathbb{Z}_3[\mu_{q+1}]$, which is the ring of integers W of the unramified extension of $\mathbb{Q}_3$ of degree $2f$ (the order of 3 modulo $q+1$ is $2f$). Let π be a uniformiser of F_u . Then $z = (\pi + i)/(\pi - i)$ and $\sigma(z) = z^{-1}$ lie in $T(F_u)$, and $2 - z - \sigma(z) = 4/(\pi^2 + 1)$ is a unit of O_{L_u} in R ; its inverse is a limit of its powers, so $(\pi^2 + 1)/4 \in R$, and $\pi = (z - \sigma(z))(\pi^2 + 1)/(4i) \in R$. Hence $R \supseteq W[\pi] = O_{L_u}$, since π is also a uniformiser of L_u . By continuity $\text{Tr}(\beta O_{L_u}) \subseteq \frac{1}{3}O_3$, so $\lambda : O_{L_u} \rightarrow \mathbb{F}_9$ is defined, $\mathbb{Z}_9$ -linear, and zero on $3O_{L_u} = \mathfrak{m}^e$.

Suppose $\lambda(\mathfrak{m}) \neq 0$, and let $j \geq 1$ be the largest integer with $\lambda(\mathfrak{m}^j) \neq 0$. Identify $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ with $\mathbb{F}_{q^2}$ through $x \mapsto (x/\pi^j) \bmod \mathfrak{m}$; then λ induces a nonzero additive map $\lambda_j : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$. For $\eta \in O_{F_u}$ the element $t = (1 + i\pi^j\eta)/(1 - i\pi^j\eta)$ lies in $T(F_u)$ (as $\sigma(t) = t^{-1}$), and $t \equiv 1 + 2i\pi^j\eta \pmod{\mathfrak{m}^{j+1}}$. For $z \in T(F_u)$ with residue $\hat{z} \in \mu_{q+1}$ this gives $\lambda(z t) = \lambda(z) + \lambda_j(2i\hat{z}\hat{\eta})$, where $\hat{\eta}$ is the residue of η . As $\hat{\eta}$ runs over $\mathbb{F}_q$, the second term runs over an additive subgroup Σ_z of $\mathbb{F}_9$, and $\lambda(z) + \Sigma_z \subseteq A'$. But $A' = \{a + bi : a, b \neq 0\}$ contains no coset of a nonzero subgroup of $\mathbb{F}_9$, so $\Sigma_z = 0$: λ_j vanishes on $i\hat{z}\mathbb{F}_q$ for every $\hat{z} \in \mu_{q+1}$. These sets span $\mathbb{F}_{q^2}$ additively, because the $\mathbb{F}_q$ -span of μ_{q+1} is a subfield not contained in $\mathbb{F}_q$; so $\lambda_j = 0$, a contradiction. Hence $\lambda(\mathfrak{m}) = 0$, the induced map λ_1 is $\mathbb{F}_9$ -linear, and $\lambda_1(\mu_{q+1}) \subseteq A'$ because $T(F_u)$ reduces onto μ_{q+1} . $\square$

Lemma 7 (residue fields). *Let f be odd and $q = 3^f$. There is an $\mathbb{F}_9$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ with $\lambda_1(\mu_{q+1}) \subseteq A'$ if and only if $f = 1$.*

Proof. For $f = 1$, $\mu_4 \subset \mathbb{F}_9$ and $\lambda_1(x) = (1 + i)x$ works. Let $f \geq 3$. Every $\mathbb{F}_9$ -linear map is $\lambda_1(x) = \text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_9}(bx) = \sum_{j < f} (bx)^{9^j}$ for some $b \in \mathbb{F}_{q^2}$. In characteristic 3,

$$\lambda_1(z)^4 = \lambda_1(z)^3 \lambda_1(z) = \sum_{j,k < f} b^{e_{jk}} z^{e_{jk}}, \quad e_{jk} = 3 \cdot 9^j + 9^k.$$

Since $A' = \{y : y^4 = -1\}$, if $\lambda_1(\mu_{q+1}) \subseteq A'$ then $\Phi(z) = 1 + \sum_{j,k} b^{e_{jk}} z^{e_{jk}}$ vanishes on μ_{q+1} . On this cyclic group, of order prime to 3, the characters $z \mapsto z^e$ ($e \bmod q+1$) are linearly independent; so for every class $e \bmod q+1$ the sum of the coefficients of the $z^{e_{jk}}$ with $e_{jk} \equiv e$ vanishes (or equals -1 , for $e \equiv 0$).

Modulo $q+1$, $g = -3$ satisfies $g^f = -3^f \equiv 1$ (as f is odd), $9^j \equiv g^{2j}$ and $3 \cdot 9^j \equiv -g^{2j+1}$. Let $e_{j'k'} \equiv e_{jj} = 4 \cdot 9^j$. Multiplying by g^{-2j} gives $4 \equiv -g^\alpha + g^\gamma$ with $\alpha \equiv 2j' + 1 - 2j$ and $\gamma \equiv 2k' - 2j \pmod{f}$, $0 \leq \alpha, \gamma < f$. As integers, $(-3)^\alpha$ and $(-3)^\gamma$ have absolute value at most 3^{f-1} , and $4 + 2 \cdot 3^{f-1} < 3^f + 1$ for $f \geq 3$; so the congruence is the equality $(-3)^\gamma - (-3)^\alpha = 4$. It forces $\gamma = 0$ (otherwise 3 divides the left side, or $(-3)^\gamma = 5$), then $(-3)^\alpha = -3$ and $\alpha = 1$; so $j' = k' = j$, as 2 is invertible modulo f . Hence the class of $4 \cdot 9^j$ contains only e_{jj} , it is not

0 (because $q + 1 > 4$ and $3 \nmid q + 1$), and its coefficient $b^{4 \cdot 9^j}$ must vanish. So $b = 0$, and then $\Phi = 1 \neq 0$. $\square$

Proof of Theorem B. If $i \in F$, then (a) and (b) fail ($F(i) = F$, and every residue field above 3 contains $\mathbb{F}_9$), and $\chi(F^2) = \infty$ by a theorem of Davies [2] (see [12, Proposition 3]); so the theorem holds. Let $i \notin F$.

Only if. Let F^2 be 3-colourable. By Lemma 4, (c) or (q) holds. In case (c), Section 4 gives (a). In case (q), Lemmas 5 and 6 give a place $u_0 \mid 3$ of odd residue degree f and an $\mathbb{F}_9$ -linear $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ with $\lambda_1(\mu_{q+1}) \subseteq A'$, and Lemma 7 gives $f = 1$: this is (b).

If. Under (a) the plane is 2-colourable [18, Theorem A'], [13, Theorem 1]. Under (b), let $u \mid 3$ have residue degree 1 and let w be the place of L above it. Then $i \notin F_u$, $L_w = F_u(i)$ has residue field $\mathbb{F}_9$, and the unit vectors of F_u^2 are units of O_w whose residues lie in μ_4 . A unit step preserves the class of z modulo O_w ; fix a representative ρ of each class and colour z by $\ell((z - \rho) \bmod \mathfrak{m}_w)$, where $\ell(a + bi) = a + b$ on $\mathbb{F}_9 = \mathbb{F}_3 + \mathbb{F}_3i$. Since $\ell(\pm 1) = \pm 1$ and $\ell(\pm i) = \pm 1$, every unit step changes the colour. This colours $F_u^2 \supseteq F^2$ with three colours. Finally $\chi(F^2) \leq 2$ if and only if (a), by [13, Theorem 1]. $\square$

The proof uses Theorem W^+ only through Lemma 2, for the finite sets G_NV ; the rest is compactness, weak approximation for the norm-one torus, the arithmetic of the places above 2 and 3, and, in case (c), the two-colour theorem of [13].

6. CONSEQUENCES

Corollary 1 (no circular chromatic number between 2 and 3). *Let F be a number field. If F^2 has a homomorphism to a circular clique $K_{p/q}$ with $p/q < 3$, then $\chi(F^2) \leq 2$. Hence $\chi_c(F^2) = 2$, $\chi_c(F^2) = 3$ or $\chi_c(F^2) > 3$, according as $\chi(F^2) = 2$, 3 or at least 4; and F^2 has a homomorphism to a cycle of odd length at least 5 only if it is bipartite.*

Proof. If $i \in F$ there is nothing to prove, so let $i \notin F$. For every finite U as in the proof of Lemma 2, Theorem W^+ gives a character with $\xi(U) \subseteq [q/p, 1 - q/p] + \mathbb{Z}$, and $q/p > 1/3$. In that proof $\varphi(v)$ then satisfies $\operatorname{Re}(\overline{\varphi(v)}) \in [q/p, 1 - q/p] + \mathbb{Z}$ (take $\gamma = 1$), while a point $N\varepsilon$ with $\varepsilon \in E_q$ (by Proposition 1 these are the points of type q) has real part in $\{\frac{1}{3}, \frac{2}{3}\} + \mathbb{Z}$, as $3 \nmid N$. So every ε_v has type c, the conditions P_v in the proof of Lemma 4 can all be taken with $\tau(v) = c$, and (c) holds; so $\chi(F^2) \leq 2$. For the second statement: if $\chi_c(F^2) < 3$, then by the definition of the infimum F^2 maps to some $K_{p/q}$ with $p/q < 3$, so $\chi(F^2) \leq 2$; hence $\chi_c(F^2) = 3$ when $\chi(F^2) = 3$, as $\chi_c \leq \chi$. If $\chi(F^2) \geq 4$, then by the theorem of de Bruijn and Erdős [4] some finite subgraph H has $\chi(H) \geq 4$, so $\chi_c(F^2) \geq \chi_c(H) > \chi(H) - 1 \geq 3$. $\square$

Call a number field F *split above 2 and 3* if $i \in F_v$ for every place v of F above 2 and above 3, that is, if every such place splits in $F(i)$; then (a) and (b) fail. The admissible fields of [13, Section 4], which must moreover satisfy $\sqrt{3} \notin F$ and $i \in F_v$ at the places above 7, 11 and 19, are split above 2 and 3.

Corollary 2. *Every number field F that is split above 2 and 3 has $\chi(F^2) \geq 4$.*

A real quadratic field $\mathbb{Q}(\sqrt{d})$ is split above 2 and 3 exactly when $d \equiv 23 \pmod{24}$, so for quadratic fields Corollary 2 is contained in [11]. Of the biquadratic fields listed as admissible in the account of [9, `notes/local_global.md`, Section 4], $\mathbb{Q}(\sqrt{2}, \sqrt{47})$, $\mathbb{Q}(\sqrt{11}, \sqrt{13})$ and $\mathbb{Q}(\sqrt{10}, \sqrt{38})$ (which contains $\sqrt{95}$) contain some $\sqrt{d}$ with $d \equiv 11 \pmod{12}$; for $\mathbb{Q}(\sqrt{2}, \sqrt{31})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{55})$ the bound is new.

Corollary 3 (local fields). *Let K be a finite extension of $\mathbb{Q}_p$. Then $\chi(K^2) = 2$ if $p = 2$ and $K(i)/K$ is ramified, $\chi(K^2) = 3$ if $p = 3$ and the residue degree of K is 1, and $\chi(K^2) \geq 4$ in every other case.*

Proof. In the first case the 2-colouring of the proof of [13, Proposition 3] colours K^2 itself (that proof colours F_v^2 , and it applies verbatim to every finite extension K of $\mathbb{Q}_2$ with $K(i)/K$ ramified); in the second case the 3-colouring of the proof of Theorem B colours K^2 itself; and $\chi(K^2) \geq 3$

when $p = 3$, because $\mathbb{Q}(\sqrt{7}) \subset \mathbb{Q}_3 \subseteq K$ and $\chi(\mathbb{Q}(\sqrt{7})^2) = 3$. If $p \geq 5$, then $\chi(K^2) \geq \chi(\mathbb{Q}_p^2) \geq 4$ by [11, Corollary 3]. If $i \in K$, then $\chi(K^2) = \infty$ [12, Proposition 3]. There remain the case $p = 2$, $i \notin K$, $K(i)/K$ unramified, and the case $p = 3$ with odd residue degree at least 3. In both we find a number field $F \subset K$ for which (a) and (b) fail; then $\chi(K^2) \geq \chi(F^2) \geq 4$ by Theorem B.

Let $m = [K : \mathbb{Q}_p]$ and let p' be the other element of $\{2, 3\}$. If $p = 2$, then m is even: if m were odd, $\mathbb{Q}_2^\times / \mathbb{Q}_2^{\times 2}$ would inject into $K^\times / K^{\times 2}$, and since -1 and -5 are not squares in $\mathbb{Q}_2$, $i \notin K$ and $K(i) \neq K(\sqrt{5})$, which is the unramified quadratic extension of K ; so $K(i)/K$ would be ramified. Choose monic polynomials over $\mathbb{Q}_p$, $\mathbb{Q}_{p'}$ and $\mathbb{Q}_5$ of a common even degree n :

- over $\mathbb{Q}_p$, $h_K h'$, where h_K is the minimal polynomial of a generator of K and h' is a product of distinct irreducible quadratics, coprime to h_K , whose roots generate $\mathbb{Q}_2(i)$ (if $p = 2$) or $\mathbb{Q}_9$ (if $p = 3$); if $p = 3$ and m is odd, h' also contains one cubic factor, coprime to h_K , generating the unramified cubic extension $\mathbb{Q}_{27}$ of $\mathbb{Q}_3$;
- over $\mathbb{Q}_{p'}$, a product of distinct irreducible quadratics whose roots generate $\mathbb{Q}_2(i)$, resp. $\mathbb{Q}_9$;
- over $\mathbb{Q}_5$, an irreducible polynomial.

By weak approximation choose a monic $h \in \mathbb{Q}[x]$ whose coefficients are so close to the three targets that, by Krasner's lemma, h factors over $\mathbb{Q}_p$, $\mathbb{Q}_{p'}$ and $\mathbb{Q}_5$ with the same degrees and the same fields as the targets. Then $F = \mathbb{Q}[x]/(h)$ is a field (as h is irreducible over $\mathbb{Q}_5$); it embeds in K , through a root of h close to the chosen generator of K ; its places above p are the place that gives K and places that contain i (or, for the cubic factor, have residue degree 3); and its places above p' all contain i . So no place above 2 ramifies in $F(i)$ and no place above 3 has residue degree 1. $\square$

For example $\chi(\mathbb{Q}_{27}^2) \geq 4$, although $i \notin \mathbb{Q}_{27}$ and $x^2 + y^2$ is anisotropic over $\mathbb{Q}_{27}$, as over $\mathbb{Q}_3$; and $\chi(\mathbb{Q}_2(\sqrt{3})^2) \geq 4$, as $\mathbb{Q}_2(\sqrt{3}, i) = \mathbb{Q}_2(\sqrt{3}, \sqrt{-3})$ is unramified over $\mathbb{Q}_2(\sqrt{3})$ (this also follows from $\chi(\mathbb{Q}(\sqrt{11})^2) = 4$, since $\mathbb{Q}_2(\sqrt{11}) = \mathbb{Q}_2(\sqrt{3})$). Corollary 3 extends [11, Corollary 3] ($p \geq 5$) and the values $\chi(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = 3$, which follow from Madore's results [17] (see [12, Theorem 2]).

7. AN EXPLICIT FAMILY: TWO SQUARE ROOTS

For many fields the general argument can be replaced by two explicit relations. For $c \geq 1$ put

$$u_c = \frac{(1 + i\sqrt{c})^2}{1 + c} = \frac{(1 - c) + 2\sqrt{c}i}{1 + c}, \quad \text{so that} \quad (1 + c)(u_c + \bar{u}_c) = 2(1 - c).$$

Proposition 2. *Let $F \subset \mathbb{R}$ contain $\sqrt{a}$ and $\sqrt{b}$, where $a \equiv 2 \pmod{3}$ and $b \equiv 7 \pmod{8}$ are positive integers ($a = b$ is allowed), and let $N = 5^k > \frac{4\sqrt{10}}{3} \max(a, b)$. Then no character of the group generated by $U = G_N\{1, u_a, \bar{u}_a, u_b, \bar{u}_b\}$ maps every element of U into $[1/3, 2/3] + \mathbb{Z}$. Hence $\chi(F^2) \geq 4$.*

Proof. As in the proof of Lemma 2, such a character gives numbers $w_v = \varphi(v) \in S_N$ ($v \in \{1, u_a, \bar{u}_a, u_b, \bar{u}_b\}$) that satisfy the two relations exactly. The sums of the absolute values of their coefficients are $4a$ and $4b$, and $6r \cdot 4c = \frac{4\sqrt{10}}{3}c < N$; so Lemma 1 gives, for the types,

$$(1 + a)(\varepsilon_{u_a} + \varepsilon_{\bar{u}_a}) = 2(1 - a)\varepsilon_1, \quad (1 + b)(\varepsilon_{u_b} + \varepsilon_{\bar{u}_b}) = 2(1 - b)\varepsilon_1.$$

At 3. $3 \mid 1 + a$ and $3 \nmid 2(1 - a)$. By (1) the left side of the first relation has $v_3 \geq 0$, hence $v_3(\varepsilon_1) \geq 0$, and ε_1 is not of type q. At 2. $v_2(1 + b) \geq 3$ and $v_2(2(1 - b)) = 2$, as $1 - b \equiv 2 \pmod{8}$. By (1) the left side of the second relation has $v_{1+i} \geq 2 \cdot 3 - 1 = 5$, while $v_{1+i}(2(1 - b)\varepsilon_1) = 4 + v_{1+i}(\varepsilon_1) = 3$ if ε_1 has type c. So ε_1 is not of type c either, a contradiction. $\square$

For $a = b = d$ this is the case $d \equiv 23 \pmod{24}$ of [11, Theorem 1], where [11, Theorem 4(1)] needs only $d < 21 \cdot 5^{k-1}$. Proposition 2 covers, for example, $\mathbb{Q}(\sqrt{2}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{31})$, $\mathbb{Q}(\sqrt{5}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{55})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ (with $b = 15$), and every field containing one of them. In terms of Theorem B, $\sqrt{a}$ makes every place above 3 have even residue degree, and $\sqrt{b}$ puts i in every completion above 2.

8. CIRCULAR COLOURINGS

For the plane over $\mathbb{R}$, DeVos, Ebrahimi, Ghebleh, Goddyn, Mohar and Naserasr proved $\chi_c(\mathbb{R}^2) \geq 4$ [6]; since de Grey's theorem gives a finite subgraph with $\chi = 5$, in fact $\chi_c(\mathbb{R}^2) > 4$. For planes over number fields the values 2 and 3 are immediate (a bipartite plane; a plane with $\chi = 3$ containing a unit triangle), and by Corollary 1 the planes with $\chi_c \leq 3$ are exactly those with $\chi \leq 3$. We know of no earlier non-integral value. Theorem C shows that $7/2$ occurs; its upper bound comes from a residue field.

Proposition 3. *Let v be a place of F whose residue field is $\mathbb{F}_p$, with $p \equiv 3 \pmod{4}$, and let $\lambda : \mathbb{F}_{p^2} \rightarrow \mathbb{F}_p$ be $\mathbb{F}_p$ -linear with $\lambda(\mu_{p+1}) \subseteq \{m, m+1, \dots, p-m\}$. Then F_v^2 , hence F^2 , has a homomorphism to $K_{p/m}$. For $p = 7$, $\lambda(a+bi) = 2a+3b$ takes the values 2, 3, 4, 5 on $\mu_8 \subset \mathbb{F}_{49}$, so F^2 maps to $K_{7/2}$, and $\chi_c(F^2) \leq 7/2$.*

Proof. As -1 is not a square in $\mathbb{F}_p$, $i \notin F_v$, and $L_w = F_v(i)$ is unramified over F_v with residue field $\mathbb{F}_{p^2}$. A unit vector z has $z\bar{z} = 1$, so it is a unit of O_w , and since the conjugation of L_w/F_v reduces to the Frobenius $x \mapsto x^p$, its residue satisfies $\hat{z}^{p+1} = 1$. A unit step preserves the class modulo O_w ; fix a representative ρ of each class and colour z by $\lambda((z - \rho) \bmod \mathfrak{m}_w) \in \mathbb{Z}/p$. A unit step changes the colour by an element of $\{m, \dots, p-m\}$. For $p = 7$, $\mu_8 = \{\pm 1, \pm i, \pm 2 \pm 2i\}$, and $2a+3b$ takes the values 2, 5, 3, 4, 3, 5, 2, 4 at $1, -1, i, -i, 2+2i, 2-2i, -2+2i, -2-2i$. $\square$

Proof of Theorem C. Upper bounds. 7 splits in $\mathbb{Q}(\sqrt{11})$ ($11 \equiv 2^2 \pmod{7}$) and ramifies in $\mathbb{Q}(\sqrt{35})$, so both fields, and $\mathbb{Q}_7$, have a place with residue field $\mathbb{F}_7$; Proposition 3 applies.

Lower bounds. For $d \in \{11, 35\}$ let $u_n = (n + i\sqrt{d})^2 / (n^2 + d)$, $V = \{1, u_1, \bar{u}_1, u_7, \bar{u}_7, u_{19}, \bar{u}_{19}\}$ and $U = G_{25}V$, a set of 140 unit vectors (70 up to sign). We checked by computer that no character of $\mathbb{Z}U$ maps every element of U into the open interval $(2/7, 5/7) + \mathbb{Z}$. The certificate branches on the integer values of the relations among the vectors (66 relations; 17 744 nodes for $d = 11$ and 14 199 for $d = 35$) and closes every leaf with an exact Farkas vector; two exact checkers that share no code accept it (Section 10). If F^2 , $F = \mathbb{Q}(\sqrt{d})$, mapped to $K_{p/q}$ with $p/q < 7/2$, Theorem W^+ would give a character with values in $[q/p, 1 - q/p] \subset (2/7, 5/7)$ on U . So $\chi_c(F^2) \geq 7/2$. Since $\mathbb{Q}(\sqrt{11}) \subset \mathbb{Q}_7$, also $\chi_c(\mathbb{Q}_7^2) \geq 7/2$.

Chromatic numbers. $K_{7/2} \rightarrow K_4$ gives $\chi \leq 4$, and $\chi \geq \chi_c = 7/2 > 3$ gives $\chi \geq 4$ (for the two quadratic fields this is also [11, Theorem 1], for $\mathbb{Q}_7$ [12, Theorem 2]). The last sentence of the theorem is Corollary 1. $\square$

The lower bound is about the infinite graph $\text{Cay}(\mathbb{Z}U, U)$, to which Theorem W^+ is applied directly; we do not know a finite unit-distance graph with circular chromatic number exactly $7/2$. The open interval is essential: the characters that come from the 7-adic colourings take values in the closed interval $[2/7, 5/7]$ on U . The certificates were found with a floating-point mixed-integer solver, which reports $\max_{\xi} \min_{u \in U} \|\xi(u)\| = 2/7$ for both sets; nothing in the proof depends on it.

The circular chromatic number separates planes with the same chromatic number:

Proposition 4. $\chi_c(\mathbb{Q}(\sqrt{59})^2) \geq 53/15 > 7/2$, while $\chi(\mathbb{Q}(\sqrt{59})^2) = 4$.

Proof. For $d = 59$ the same set $U = G_{25}V$ has no character into the open interval $(15/53, 38/53)$: the certificate has 127 181 nodes and 105 761 leaves, and both checkers accept it (Section 10). As in the proof of Theorem C, Theorem W^+ gives $\chi_c \geq 53/15$. Here 7 is inert, so Proposition 3 does not apply. We have $\chi \geq 4$ by [11, Theorem 1], as $59 \equiv 11 \pmod{12}$. And $\chi \leq 4$: the place of $F = \mathbb{Q}(\sqrt{59})$ above 2 has residue field $\mathbb{F}_2$ and is inert in $F(i)$ (as $59 \equiv 3 \pmod{8}$), so the unit vectors reduce into the group μ_3 of elements of norm one in $\mathbb{F}_4$, and, as in the proof of Proposition 3, the residue modulo the maximal ideal is a proper 4-colouring (a bound of Fischer [7]). $\square$

Corollary 4. $\chi_c(E^2) = 7/2$ for every finite extension E of $\mathbb{Q}_7$ with residue degree 1, and $\chi_c(F^2) = 7/2$ for every number field F that contains $\sqrt{11}$ or $\sqrt{35}$ and has a place above 7 of residue degree 1, for example $\mathbb{Q}(\sqrt{7}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{35})$.

Proof. $E \supseteq \mathbb{Q}_7 \supset \mathbb{Q}(\sqrt{11})$ and $F \supseteq \mathbb{Q}(\sqrt{11})$ or $\mathbb{Q}(\sqrt{35})$ give $\chi_c \geq 7/2$ by Theorem C; the residue field $\mathbb{F}_7$ gives $\chi_c \leq 7/2$ by Proposition 3 (whose proof applies to E itself). $\square$

Locally constant colourings. The values 2, 3 and $7/2$ are the only ones below 4 that colourings through a single completion can give (Corollary 5). Let E be a finite extension of $\mathbb{Q}_p$ with $i \notin E$ and $E(i)/E$ unramified (for p odd this always holds; for $p = 2$ it is the case where (a) fails), let $\mathbb{F}_q$ be the residue field of E ($q = p^f$), w the place of $L = E(i)$, and

$$\kappa_1(E) = \max_{\lambda} \min_{z \in \mu_{q+1}} \|\lambda(z)\|,$$

the maximum over the additive maps $\lambda : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$.

Proposition 5. *Suppose E^2 has a locally constant homomorphism to $K_{P/Q}$, $P/Q < 4$. Then $Q/P \leq \kappa_1(E)$. Conversely E^2 maps to $K_{p/m}$ when $m/p = \kappa_1(E) > 0$.*

Proof. The converse is the colouring of Proposition 3, with $\mathbb{F}_q$ in place of $\mathbb{F}_p$. Let π be a uniformiser of E (also one of L) and σ the non-trivial automorphism of L/E . On the compact set O_w the homomorphism is constant on the cosets of some power $\mathfrak{m}^k$ of the maximal ideal, and only its restriction to O_w is used. It induces a homomorphism from $\text{Cay}(O_w/\mathfrak{m}^k, \bar{T})$, where $\bar{T}$ is the image of $T(E)$, to $K_{P/Q}$; by Theorem W⁺ there is a character ξ of $O_w/\mathfrak{m}^k$ with $\|\xi(z)\| \geq Q/P > 1/4 \geq 1/(2p)$ for every $z \in T(E)$. We show that ξ vanishes on $\mathfrak{m}$; then ξ is a map λ as in the definition of κ_1 , and $Q/P \leq \kappa_1(E)$, since $T(E)$ reduces onto μ_{q+1} (it contains the Teichmüller lifts of μ_{q+1} , on which σ acts as $x \mapsto x^q$). Suppose not, and let $J \geq 1$ be the largest integer with $\xi(\mathfrak{m}^J) \neq 0$. Through $x \mapsto (x/\pi^J) \bmod \mathfrak{m}$, ξ induces a nonzero additive map $\lambda_J : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$. For $x \in O_w$, $t = (1 + \pi^J x)/\sigma(1 + \pi^J x)$ lies in $T(E)$ and $t \equiv 1 + \pi^J(x - \sigma(x)) \pmod{\mathfrak{m}^{J+1}}$, and the residues of $x - \sigma(x)$ fill the $\mathbb{F}_q$ -line $D = \{y - y^q : y \in \mathbb{F}_{q^2}\}$. For $z \in T(E)$ with residue $\hat{z}$, $\xi(zt) = \xi(z) + \lambda_J(\hat{z}d)$, $d \in D$; the values $\lambda_J(\hat{z}D)$ form a subgroup of $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$, and if it were the whole group, $\xi(T(E))$ would contain a coset of $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$ and a point with $\|\cdot\| \leq \frac{1}{2p}$. So λ_J vanishes on $\hat{z}D$ for every $\hat{z} \in \mu_{q+1}$, hence on their additive span, which is D times the $\mathbb{F}_q$ -span of μ_{q+1} , that is $\mathbb{F}_{q^2}$: a contradiction. $\square$

Proposition 6. $\kappa_1(E) > 1/4$ only when $p = 3$ and $f = 1$, where $\kappa_1 = 1/3$, or $p = 7$ and $f = 1$, where $\kappa_1 = 2/7$. In every other case with $i \notin E$ and $E(i)/E$ unramified, $\kappa_1(E) < 1/4$; it is 0 when $p = 2$ or $f \geq 3$.

Proof. For $p = 2$: if $\lambda(\mu_{q+1}) = \{\frac{1}{2}\}$, summing over μ_{q+1} , whose elements add up to 0, gives $\lambda(0) = \frac{q+1}{2} \equiv \frac{1}{2}$, as $q+1$ is odd; so some z has $\lambda(z) = 0$.

Let p be odd; then $p \equiv 3 \pmod{4}$ and f is odd, as $i \notin E$. Let $\psi(x) = e^{2\pi i x/p}$ on $\mathbb{F}_p$, and $S(b) = \sum_{z \in \mu_{q+1}} \psi(\text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_p}(bz))$ for $b \in \mathbb{F}_{q^2}^\times$. Every additive $\lambda : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$ is $z \mapsto \frac{1}{p} \text{Tr}(bz)$ for some b . Writing the indicator function of $\mu_{q+1} = \ker N_{\mathbb{F}_{q^2}/\mathbb{F}_q}$ with the multiplicative characters χ of $\mathbb{F}_q^\times$, and using the Davenport–Hasse relation $G(\chi \circ N, \psi \circ \text{Tr}) = -G(\chi, \psi_q)^2$ for Gauss sums [3] ($\psi_q = \psi \circ \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}$; see [16, Chapter 5]), one finds

$$S(b) = \frac{1}{q-1} \sum_{\chi} \bar{\chi}(N(b)) G(\chi \circ N, \psi \circ \text{Tr}) = - \sum_{y \in \mathbb{F}_q^\times} \psi_q\left(y + \frac{N(b)}{y}\right),$$

minus a Kloosterman sum; so $|S(b)| \leq 2\sqrt{q}$ by Weil’s bound [21]. Hence, for $b \neq 0$, the number of $z \in \mu_{q+1}$ with $\text{Tr}(bz) = 0$ is $\frac{1}{p} \sum_{a \in \mathbb{F}_p} \sum_z \psi(a \text{Tr}(bz)) = \frac{q+1}{p} + \frac{1}{p} \sum_{a \neq 0} S(ab) \geq \frac{q+1}{p} - \frac{p-1}{p} 2\sqrt{q}$, which is positive when $f \geq 3$ (then $\sqrt{q} \geq p^{3/2} > 2(p-1)$); so $\kappa_1 = 0$ for $f \geq 3$.

Let $f = 1$, $p = 4k + 3$, and $I = \{x : |x| \leq k\} \subset \mathbb{F}_p$, the residues with $\|x/p\| < \frac{1}{4}$. The number of $z \in \mu_{p+1}$ with $\text{Tr}(bz) \in I$ is

$$\frac{(p+1)|I|}{p} + \frac{1}{p} \sum_{a \neq 0} \left(\sum_{x \in I} \psi(-ax) \right) S(ab) \geq \frac{p^2-1}{2p} - \frac{2\sqrt{p}}{p} \sum_{a=1}^{p-1} \frac{1}{|\sin(\pi a/p)|} \geq \frac{p^2-1}{2p} - 2\sqrt{p}(\ln p + 1),$$

because $|\sum_{x \in I} \psi(-ax)| = |\sin(\pi a(2k+1)/p)/\sin(\pi a/p)| \leq 1/|\sin(\pi a/p)|$, and, by the symmetry $a \leftrightarrow p-a$ and $\sin x \geq 2x/\pi$ on $[0, \pi/2]$, $\sum_{a=1}^{p-1} 1/|\sin(\pi a/p)| \leq p \sum_{a \leq (p-1)/2} 1/a \leq p(\ln p + 1)$. The last expression is positive for $p = 1001$ and increasing for $p \geq 1001$ (its derivative is $\frac{1}{2} + \frac{1}{2p^2} - \frac{\ln p + 3}{\sqrt{p}} > 0$ there). For the primes $p \equiv 3 \pmod{4}$ with $11 \leq p < 3000$ a direct computation (in $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$, where bz runs over the circle $s^2 + t^2 = N(b)$ and $\text{Tr}(bz) = 2s$) gives $\kappa_1 < \frac{1}{4}$; the largest value is $\frac{4}{19}$. For $p = 3$ and $p = 7$ a direct computation gives $\kappa_1 = \frac{1}{3}$ and $\kappa_1 = \frac{2}{7}$, attained by the maps of the proof of Theorem B and of Proposition 3. $\square$

Corollary 5 (local circular values). *Let E be a finite extension of $\mathbb{Q}_p$. If E^2 has a locally constant homomorphism to $K_{P/Q}$ with $P/Q < 4$, then either $p = 2$ and $E(i)/E$ is ramified, or $(p, f) = (3, 1)$ and $P/Q \geq 3$, or $(p, f) = (7, 1)$ and $P/Q \geq 7/2$.*

Proof. If $i \in E$, then $\chi(E^2) = \infty$ [12, Proposition 3]. Otherwise combine Propositions 5 and 6. $\square$

So the restriction to F^2 of a locally constant colouring of a completion F_v^2 of a number field gives, below 4, only the bounds 2 (condition (a)), 3 (condition (b)) and $7/2$ (a place above 7 of residue degree 1).

9. A GAP ABOVE 3

By Corollary 1, $\chi_c(F^2)$ is 2, 3 or larger than 3. Theorem D says that in the last case it stays away from 3. Its proof is that of Theorem B with $I = [1/3, 2/3]$ replaced by a slightly wider interval; the new ingredient is the shape of the characters along the probe for such intervals.

Notation. In this section $N = 5^k$ with $k \geq 1$, $\rho = (3 + 4i)/5$, $h = (1 + i)/2$ and $\Lambda_{\pm} = \frac{2 \pm i}{5} \mathbb{Z}[i]$. For $\frac{1}{4} < \theta \leq \frac{1}{3}$ let S_N^θ be defined as S_N with $[\theta, 1 - \theta]$ in place of I , and put

$$s = \frac{1}{2} - \theta, \quad \delta = \frac{1}{3} - \theta, \quad B_s = \{a + bi : a, b \in \mathbb{R}, |a|, |b| \leq s\},$$

so that $s = \frac{1}{6} + \delta$, and $P_k = \{x \in \mathbb{C} : \bar{x}\rho^j \in B_s \text{ for } |j| \leq k\}$. For $\varepsilon \in E_q$ and $|j| \leq k$, Lemma 8 below gives $\eta_j^N(\varepsilon) \in \{\pm 1 \pm i\}$ with $\overline{N\varepsilon}\rho^j \in h + \eta_j^N(\varepsilon)/6 + \mathbb{Z}[i]$; for $1 \leq m \leq k$ let

$$Y_m^N(\varepsilon) = \{y \in \mathbb{C} : \eta_j^N(\varepsilon)/6 + \bar{y}\rho^j \in B_s \text{ for } |j| \leq m\},$$

and $Y_k(\varepsilon) = Y_k^N(\varepsilon)$. These depend on θ , on N , and on ε only modulo $\mathbb{Z}[i]$; when N is fixed we write $\eta_j(\varepsilon)$ for $\eta_j^N(\varepsilon)$. We write $w_1 = \text{Re } w$ and $w_2 = \text{Im } w$.

Lemma 8. (i) $G_N = \{i^a \rho^j : a \in \mathbb{Z}, |j| \leq k\}$, and $c \in S_N^\theta$ if and only if $\bar{c}\rho^j \in h + B_s + \mathbb{Z}[i]$ for every $|j| \leq k$.

(ii) If $\varepsilon \in E_c$ and $|j| \leq k$, then $\overline{N\varepsilon}\rho^j \in h + \mathbb{Z}[i]$.

(iii) If $\varepsilon = \frac{\alpha + \beta i}{3} + t \in E_q$ and $|j| \leq k$, then $\overline{N\varepsilon}\rho^j \equiv (-1)^k (-i)^j \frac{\alpha - \beta i}{3} \pmod{\mathbb{Z}[i]}$. This lies in $h + \eta/6 + \mathbb{Z}[i]$ for a unique $\eta \in \{\pm 1 \pm i\}$; moreover $\eta_{j-1}(\varepsilon) = i\eta_j(\varepsilon)$ for $1 - k \leq j \leq k$ (so $\eta_{j-2}(\varepsilon) = -\eta_j(\varepsilon)$), and $\eta_j(\bar{\varepsilon}) = \overline{\eta_{-j}(\varepsilon)}$.

(iv) $N\varepsilon + P_k \subseteq S_N^\theta$ for $\varepsilon \in E_c$, and $N\varepsilon + Y_k(\varepsilon) \subseteq S_N^\theta$ for $\varepsilon \in E_q$.

Proof. (i) If $\gamma \in G_N$, then $N\gamma \in \mathbb{Z}[i]$ has norm 5^{2k} , so $N\gamma = u(2+i)^a(2-i)^b$ with a unit u and $a+b=2k$, and $\gamma = u(\frac{2+i}{2-i})^{a-k} = u\rho^{a-k}$; conversely $N\rho^{\pm j} = 5^{k-j}(3 \pm 4i)^j \in \mathbb{Z}[i]$ for $0 \leq j \leq k$. Since $\text{Re}(\pm iw) = \mp \text{Im } w$ and $[\theta, 1 - \theta] + \mathbb{Z} = \frac{1}{2} + [-s, s] + \mathbb{Z}$ is symmetric, $\text{Re}(\bar{c}\gamma) \in [\theta, 1 - \theta] + \mathbb{Z}$ for all $\gamma \in G_N$ if and only if both coordinates of every $\bar{c}\rho^j$ lie in $\frac{1}{2} + [-s, s] + \mathbb{Z}$.

(ii) Write $\varepsilon = h + t$. The Gaussian integer $N\rho^j$ has odd norm, so $N\rho^j = 1 + (1+i)m$ with $m \in \mathbb{Z}[i]$, and $\bar{h}N\rho^j = \bar{h} + m \equiv h$, as $\bar{h} = h - i$; and $\bar{t}N\rho^j \in \mathbb{Z}[i]$.

(iii) $\overline{N\varepsilon}\rho^j \equiv \frac{\alpha - \beta i}{3} N\rho^j$, and $N\rho^j \equiv (-1)^k (-i)^j \pmod{3}$, because $5 \equiv -1$ and $3 \pm 4i \equiv \pm i \pmod{3}$. Modulo $\mathbb{Z}[i]$, $\frac{\alpha - \beta i}{3}$ times a unit has both coordinates in $\{\frac{1}{3}, \frac{2}{3}\}$, so it lies in $h + \frac{\pm 1 \pm i}{6} + \mathbb{Z}[i]$. Since $(-i)^{j-1} = i(-i)^j$ and $ih \equiv h$, $\eta_{j-1} = i\eta_j$. Finally $\overline{N\varepsilon}\rho^j$ is the conjugate of $\overline{N\varepsilon}\rho^{-j}$, and $\bar{h} \equiv h$.

(iv) follows from (i)–(iii). $\square$

Lemma 9. (i) $P_1 = sP$, where P is the 12-gon with vertices $\pm 1 \pm \frac{i}{3}$, $\pm \frac{1}{3} \pm i$ and $\frac{5}{7}(\pm 1 \pm i)$; so $|x| \leq s\sqrt{10}/3$ for $x \in P_k$.

(ii) $|y| \leq \sqrt{2}\delta$ for $y \in Y_m^N(\varepsilon)$, $1 \leq m \leq k$.

Proof. (i) P_1 is defined by the twelve inequalities $\pm(\bar{x}\rho^j)_e \leq s$, $j \in \{0, \pm 1\}$, $e \in \{1, 2\}$, which are homogeneous in (x, s) ; intersecting their boundary lines in pairs gives the vertices. For instance at $x = 1 + \frac{i}{3}$ we have $\bar{x} = 1 - \frac{i}{3}$, $\bar{x}\rho = \frac{13}{15} + \frac{3}{5}i$ and $\bar{x}\rho^{-1} = \frac{1}{3} - i$. Every vertex has modulus at most $\sqrt{10}/3$, and $P_k \subseteq P_1$.

(ii) As $s = \frac{1}{6} + \delta$, the condition $\eta_j/6 + \bar{y}\rho^j \in B_s$ implies $\eta_{j,e}(\bar{y}\rho^j)_e \leq \delta$ for $e = 1, 2$, where $\eta_j = \eta_{j,1} + \eta_{j,2}i$. So $Y_m^N(\varepsilon) \subseteq Y_1^N(\varepsilon) \subseteq \delta X(\varepsilon)$, where $X(\varepsilon)$ is defined by the six inequalities $\eta_{j,e}(\bar{y}\rho^j)_e \leq 1$, $|j| \leq 1$ (for $\delta = 0$ read $\delta X(\varepsilon)$ as $\{0\}$: as $X(\varepsilon)$ is bounded, the same inequalities with right-hand side 0 have only the solution 0). By Lemma 8(iii), $\eta_j^N(\varepsilon) = \eta_j^N(\pm\varepsilon)$ for $|j| \leq 1$; replacing ε by $i\varepsilon$ replaces η_j by $-i\eta_j$ and X by iX , and multiplication by i permutes the four classes of E_q modulo $\mathbb{Z}[i]$ cyclically. So every $X(\varepsilon)$ is a rotation of $X(\frac{1+i}{3})$ at level 5, where $\eta_{-1} = 1 + i$, $\eta_0 = 1 - i$, $\eta_1 = -1 - i$. This is the hexagon with vertices $-\frac{5}{4}$, $-\frac{5}{4}i$, $-\frac{5}{7}(1 + i)$, $1 - \frac{i}{2}$, $1 + i$ and $-\frac{1}{2} + i$, of moduli at most $\sqrt{2}$. $\square$

Lemma 10. Let $0 < s < 11/56$ and $k \geq 1$, and let $x \in \mathbb{C}$ with $|x| \leq s\sqrt{2}$ and $\bar{x}\rho^{k-2} \in B_s$. Then $\bar{x}\rho^k + \nu \notin B_s + \mathbb{Z}[i]$ for every $\nu \in \Lambda_+ \setminus \mathbb{Z}[i]$. The same holds with ρ^{-k} , ρ^{2-k} and Λ_- in place of ρ^k , ρ^{k-2} and Λ_+ .

Proof. Let $u = \bar{x}\rho^k$ and $w = \bar{x}\rho^{k-2} = u\rho^{-2} = u(-7 - 24i)/25$, so that $w_1 = (-7u_1 + 24u_2)/25$ and $w_2 = -(24u_1 + 7u_2)/25$. Suppose $u + \nu \in B_s + \mathbb{Z}[i]$; changing ν by a Gaussian integer we may assume $u + \nu \in B_s$. The non-zero classes of $\Lambda_+/\mathbb{Z}[i]$ are represented by $\pm \frac{2+i}{5}$ and $\pm \frac{-1+2i}{5}$, whose coordinates have absolute value at most $\frac{2}{5}$; every other element of the class has a coordinate of absolute value at least $\frac{3}{5}$, which $u_e + \nu_e \in [-s, s]$ excludes, as $|u_e| \leq |x| \leq s\sqrt{2}$ and $s(1 + \sqrt{2}) < \frac{3}{5}$. If $\nu = \pm \frac{2+i}{5}$, then $\mp u_1 \geq \frac{2}{5} - s$ and $\mp u_2 \geq \frac{1}{5} - s$, so $\pm w_2 = \mp(24u_1 + 7u_2)/25 \geq (11 - 31s)/25$. If $\nu = \pm \frac{-1+2i}{5}$, then $\pm u_1 \geq \frac{1}{5} - s$ and $\mp u_2 \geq \frac{2}{5} - s$, so $\mp w_1 = \pm(7u_1 - 24u_2)/25 \geq (11 - 31s)/25$. In both cases $|w_e| \geq (11 - 31s)/25 > s$, as $56s < 11$, contradicting $w \in B_s$. The second statement follows by complex conjugation, which exchanges ρ and ρ^{-1} , and Λ_+ and Λ_- , and preserves B_s and $\mathbb{Z}[i]$ (apply the first to $\bar{x}$ and $\bar{\nu}$). $\square$

Lemma 11. Let $0 \leq \delta < 5/168$, $k \geq 2$ and $\varepsilon \in E_q$, and let $y \in \mathbb{C}$ with $|y| < \frac{1}{6}$ and $\eta_{k-2}(\varepsilon)/6 + \bar{y}\rho^{k-2} \in B_s$. Then $\overline{N\varepsilon + y}\rho^k + \nu \notin h + B_s + \mathbb{Z}[i]$ for every $\nu \in \Lambda_+ \setminus \mathbb{Z}[i]$. The same holds with ρ^{-k} , ρ^{2-k} , $\eta_{2-k}(\varepsilon)$ and Λ_- in place of ρ^k , ρ^{k-2} , $\eta_{k-2}(\varepsilon)$ and Λ_+ .

Proof. Replacing (ε, y, ν) by $(i\varepsilon, iy, -i\nu)$ multiplies $\overline{N\varepsilon + y}\rho^k + \nu$ by $-i$ and replaces η_j by $-i\eta_j$; it preserves $h + B_s + \mathbb{Z}[i]$ (as $-ih \equiv h$), E_q , Λ_+ and the hypotheses. So we may assume $\eta_k(\varepsilon) = 1 + i$, and then $\eta_{k-2}(\varepsilon) = -1 - i$ (Lemma 8). Let $u = \bar{y}\rho^k$, so that $\bar{y}\rho^{k-2} = u(-7 - 24i)/25$. Suppose the conclusion fails. Then $\frac{1}{6} + \nu_e + u_e \in [-s, s] + \mathbb{Z}$ for $e = 1, 2$, where we take $\nu_e \in \{\pm \frac{1}{5}, \pm \frac{2}{5}\}$. As $|u_e| < \frac{1}{6}$ and $s < \frac{1}{5}$, the values $\nu_e = \frac{1}{5}$ and $\nu_e = \frac{2}{5}$ are impossible ($\frac{1}{6} + \nu_e + u_e$ then lies in $(\frac{1}{5}, \frac{11}{15})$), and $\nu_e = -\frac{2}{5}$ forces $u_e \geq \frac{1}{15} - \delta$. Of the four non-zero classes $\pm(\frac{2}{5}, \frac{1}{5})$, $\pm(-\frac{1}{5}, \frac{2}{5})$ of $\Lambda_+/\mathbb{Z}[i]$ only $\nu = (-\frac{2}{5}, -\frac{1}{5})$ remains, so $u_1 \geq \frac{1}{15} - \delta$. On the other hand the hypothesis at ρ^{k-2} says that both coordinates of $u(-7 - 24i)/25$ are at least $-\delta$: $7u_1 - 24u_2 \leq 25\delta$ and $24u_1 + 7u_2 \leq 25\delta$. Adding 7 times the first to 24 times the second gives $625u_1 \leq 775\delta$, that is $u_1 \leq \frac{31}{25}\delta$. Hence $\frac{1}{15} \leq \frac{56}{25}\delta$, that is $\delta \geq \frac{5}{168}$, a contradiction. The second statement follows by complex conjugation, as in Lemma 10: $\bar{\varepsilon} \in E_q$, $\bar{h} \equiv h$, and $\eta_{k-2}(\bar{\varepsilon}) = \overline{\eta_{2-k}(\varepsilon)}$ by Lemma 8(iii). $\square$

Proposition 7. Let $17/56 < \theta \leq 1/3$, $k \geq 1$ and $N = 5^k$. Then

$$S_N^\theta = \bigcup_{\varepsilon \in E_c} (N\varepsilon + P_k) \cup \bigcup_{\varepsilon \in E_q} (N\varepsilon + Y_k(\varepsilon)).$$

In particular every $c \in S_N^\theta$ is $c = N\varepsilon + x$, either with $\varepsilon \in E_c$ and $|x| \leq s\sqrt{10}/3$, or with $\varepsilon \in E_q$ and $|x| \leq \sqrt{2}\delta$.

For $\theta = 1/3$ we have $s\sqrt{10}/3 = \sqrt{10}/18$ and $\delta = 0$, and this is Proposition 1.

Proof. The inclusion $\supseteq$ is Lemma 8(iv). Note that $\theta > 17/56$ means $s < 11/56$ and $\delta < 5/168$, and that then $s(1 + \sqrt{2}) < \frac{3}{5}$. The sets on the right are invariant under $N\mathbb{Z}[i]$ and under multiplication by i (which maps E_c and E_q to themselves modulo $\mathbb{Z}[i]$), and so is S_N^θ . We use induction on k . For $\lambda \in \mathbb{Z}[i]$ put $\nu_\pm(\lambda) = \frac{N}{5}\bar{\lambda}\rho^{\pm k}$. Since $\frac{N}{5}\rho^k = \frac{(2+i)^{2k}}{5} = \frac{(2+i)^{2k-1}}{2-i}$, we have $\nu_\pm(\lambda) \in \Lambda_\pm$, and $\nu_+(\lambda) \in \mathbb{Z}[i]$ if and only if $2+i$ divides λ ; likewise $\nu_-(\lambda) \in \mathbb{Z}[i]$ if and only if $2-i$ divides λ . So both are Gaussian integers if and only if $5 \mid \lambda$.

The case $k = 1$. Let $c \in S_5^\theta$. The condition at ρ^0 gives $c \in h + B_s + \mathbb{Z}[i]$, so $c = 5h + x + \lambda$ with $x \in B_s$ and $\lambda \in \mathbb{Z}[i]$, which we take modulo 5; put $z = \bar{x}$, so $|z| \leq s\sqrt{2}$. By Lemma 8(ii) the conditions at $\rho^{\pm 1}$ read $z\rho^{\pm 1} + \nu_\pm(\lambda) \in B_s + \mathbb{Z}[i]$, and as in the proof of Lemma 10 the shift can be taken to be the representative of $\nu_\pm(\lambda)$ modulo $\mathbb{Z}[i]$ with coordinates in $[-\frac{2}{5}, \frac{2}{5}]$. If both $\nu_\pm(\lambda)$ are Gaussian integers, then $5 \mid \lambda$ and $x \in P_1$ (as $|z\rho^{\pm 1}| \leq s\sqrt{2} < 1 - s$). If exactly one is, say $\nu_-(\lambda)$, then $z\rho^{-1} \in B_s$, and Lemma 10 with $k = 1$ contradicts the condition at ρ ; the other case is symmetric. In the remaining 16 classes, multiplication by i leaves four orbits, represented by $\lambda = 1, 2, 2+2i$ and $1+i$. Write $(z\rho)_{1,2} = \frac{1}{5}(3z_1 - 4z_2, 4z_1 + 3z_2)$ and $(z\rho^{-1})_{1,2} = \frac{1}{5}(3z_1 + 4z_2, -4z_1 + 3z_2)$.

- $\lambda = 1$: $\nu_\pm \equiv \frac{-2 \mp i}{5}$, so $(z\rho)_1, (z\rho^{-1})_1 \geq \frac{2}{5} - s$. Their sum is $\frac{6}{5}z_1 \leq \frac{6}{5}s$, so $\frac{4}{5} - 2s \leq \frac{6}{5}s$ and $s \geq \frac{1}{4}$.
- $\lambda = 2$: $\nu_\pm \equiv \frac{1 \mp 2i}{5}$, so $(z\rho)_1, (z\rho^{-1})_1 \leq s - \frac{1}{5}$ and $(z\rho)_2 \geq \frac{2}{5} - s$. As $7(z\rho)_1 + 25(z\rho^{-1})_1 - 24(z\rho)_2 = 0$, we get $0 \leq 32(s - \frac{1}{5}) + 24(s - \frac{2}{5})$, so $s \geq \frac{2}{7}$.
- $\lambda = 2 + 2i$: $\nu_+ \equiv \frac{-1+2i}{5}$ and $\nu_- \equiv \frac{-2+i}{5}$, so $(z\rho)_2 \leq s - \frac{2}{5}$, $(z\rho^{-1})_1 \geq \frac{2}{5} - s$ and $(z\rho^{-1})_2 \leq s - \frac{1}{5}$. As $25(z\rho)_2 - 24(z\rho^{-1})_1 + 7(z\rho^{-1})_2 = 0$, we get $0 \leq 49(s - \frac{2}{5}) + 7(s - \frac{1}{5})$, so $s \geq \frac{3}{8}$.
- $\lambda = 1 + i$: then $c = 5\varepsilon + y$ with $\varepsilon = \frac{2+2i}{3} \in E_q$ and $y = x + \frac{1+i}{6}$, so $|y| \leq \sqrt{2}(s + \frac{1}{6}) < 0.52$. By Lemma 8(iii), $\bar{c}\rho^j \in h + \eta_j(\varepsilon)/6 + \bar{y}\rho^j + \mathbb{Z}[i]$, and as $|\bar{y}\rho^j| < 0.52$ and $\frac{1}{6} + 0.52 < 1 - s$, the condition at ρ^j says $\eta_j(\varepsilon)/6 + \bar{y}\rho^j \in B_s$. So $y \in Y_1^5(\varepsilon)$.

The first three orbits are thus empty, as $s < 11/56 < \frac{1}{4}$.

The step from $k-1$ to $k \geq 2$. Let $c \in S_N^\theta$. As $G_{N/5} \subseteq G_N$, $c \in S_{N/5}^\theta$, and by induction $c = \frac{N}{5}\varepsilon' + x$ with $\varepsilon' \in E_c$ and $x \in P_{k-1}$, or with $\varepsilon' \in E_q$ and $x \in Y_{k-1}^{N/5}(\varepsilon')$.

If $\varepsilon' = h + t \in E_c$, then $\frac{N}{5}\varepsilon' = Nh + \frac{N}{5}(t - 2 - 2i)$, so $c = Nh + x + \frac{N}{5}\lambda$ with $\lambda \in \mathbb{Z}[i]$. By Lemma 8(ii) the condition at $\rho^{\pm k}$ reads $\bar{x}\rho^{\pm k} + \nu_\pm(\lambda) \in B_s + \mathbb{Z}[i]$. Here $|x| \leq s\sqrt{10}/3 < s\sqrt{2}$ by Lemma 9, and $\bar{x}\rho^{\pm(k-2)} \in B_s$ as $x \in P_{k-1}$. By Lemma 10 both $\nu_\pm(\lambda)$ are Gaussian integers, so $5 \mid \lambda$ and $c \in Nh + x + N\mathbb{Z}[i]$; and $\bar{x}\rho^{\pm k} \in B_s + \mathbb{Z}[i]$ with $|x| < 1 - s$ gives $x \in P_k$.

If $\varepsilon' = \frac{\alpha' + \beta'i}{3} + t' \in E_q$, let $\alpha, \beta \in \{1, 2\}$ with $\alpha \equiv 2\alpha'$ and $\beta \equiv 2\beta' \pmod{3}$, and $\varepsilon = \frac{\alpha + \beta i}{3}$. Then $\frac{N}{5}\varepsilon' - N\varepsilon \in \frac{N}{5}\mathbb{Z}[i]$, because $\alpha' - 5\alpha \equiv \alpha' - 2\alpha \equiv 0 \pmod{3}$, and likewise for β ; so $c = N\varepsilon + x + \frac{N}{5}\lambda$ with $\lambda \in \mathbb{Z}[i]$. For $|j| \leq k-1$ we have $\frac{N}{5}\rho^j \in \mathbb{Z}[i]$, so $\eta_j^N(\varepsilon) = \eta_j^{N/5}(\varepsilon')$ for $|j| \leq k-1$, and $x \in Y_{k-1}^N(\varepsilon)$. In particular $|x| \leq \sqrt{2}\delta < \frac{1}{6}$, and the hypotheses of Lemma 11 hold at $\rho^{\pm(k-2)}$. The condition at $\rho^{\pm k}$ says $\overline{N\varepsilon + x}\rho^{\pm k} + \nu_\pm(\lambda) \in h + B_s + \mathbb{Z}[i]$, so by Lemma 11 both $\nu_\pm(\lambda)$ are Gaussian integers, $5 \mid \lambda$ and $c \in N\varepsilon + x + N\mathbb{Z}[i]$. Finally $\eta_{\pm k}(\varepsilon)/6 + \bar{x}\rho^{\pm k} \in B_s + \mathbb{Z}[i]$ with $|x| < \frac{1}{2}$ gives $x \in Y_k(\varepsilon)$. The bounds on $|x|$ are Lemma 9. $\square$

Proof of Theorem D. Suppose that F^2 has a homomorphism to $K_{p/q}$ with $p/q < 56/17$; we show that $\chi(F^2) \leq 3$. If $i \in F$ this cannot happen, as then $\chi(F^2) = \infty$ (proof of Theorem B). If $p/q \leq 3$, then $\chi(F^2) \leq \chi(K_{p/q}) \leq 3$. Otherwise $\theta = q/p \in (17/56, 1/3)$, and we follow the proof of Lemma 2 for a finite $V \subset T$. Theorem W⁺ applies, as $p < 4q$, and gives a character with $\xi(U) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$; so $\varphi(v) \in S_N^\theta$ for $v \in V$, and Proposition 7 writes $\varphi(v) = N\varepsilon_v + x_v$ with $\varepsilon_v \in E$ and $|x_v| \leq r_\theta = \max(s\sqrt{10}/3, \sqrt{2}\delta)$ (so $r_{1/3} = r$). Lemma 1 and its proof hold for such points with r_θ in place of r . So, choosing $N > 6r_\theta \max_v(|D_v| + \sum_j |\mu_{v,j}|)$, we obtain $\beta \in L$ with $\text{Tr}(\beta v) \in E$ for every $v \in V$: the conclusion of Lemma 2. The proof of Lemma 4 uses the 3-colouring only through this conclusion, and the rest of the proof of Theorem B only through (c) or (q). So (a) or (b) holds, and $\chi(F^2) \leq 3$ by Theorem B. Finally, if $\chi(F^2) \geq 4$ and

$\chi_c(F^2) < 56/17$, the definition of χ_c as an infimum gives a homomorphism to some $K_{p/q}$ with $p/q < 56/17$, which we have excluded.

For the second statement let $\theta^* = 0.3001611$, with s^*, δ^* its values of s, δ . It suffices that for every $k \geq 17$ the conclusion of Proposition 7 holds for θ^* , so that every $c \in S_N^{\theta^*}$ is $N\varepsilon + x$ with $\varepsilon \in E$ and $|x| \leq r_{\theta^*} < 0.22$: then for $p/q < 1/\theta^*$ the argument above applies with $\theta = q/p$, since $S_N^\theta \subseteq S_N^{\theta^*}$, with r_{θ^*} in place of r_θ and $k \geq 17$. This is the computer-assisted step. We computed $S_N^{\theta^*}$ exactly, as a finite union of rational polygons modulo N , level by level, with three programs written separately (two of them by referees): all three find the same number of components at every level $k \leq 19$, and only the components of Proposition 7 for $k = 17, 18, 19$. For $k \geq 18$ the induction step of Proposition 7 goes through with Lemmas 10 and 11 replaced by the same statements under the hypotheses $x \in P_{k-1}$, resp. $y \in Y_{k-1}^N(\varepsilon)$, for every shift $\nu \in \Lambda_\pm \setminus \mathbb{Z}[i]$ (the programs do not use the reduction to the shifts of the hand proofs). The infimum of the θ for which these statements hold is an exact linear-programming value. It is non-increasing in k : in the coordinates $\bar{x}\rho^k$, resp. $\bar{y}\rho^k$, and since $\eta_{k-m} = i^m\eta_k$ (Lemma 8), the system at level $k+1$ contains the one at level k . All three programs find that it is at most $0.30005568 < \theta^*$ for $k = 12, 13$ and 14 (two of them up to $k = 22$). $\square$

Remark. (1) For a statement valid for every k , the bound $17/56$ in Proposition 7 is best possible: for $\theta = 17/56$, S_N^θ has further isolated points for $N = 5, 25, 125$ and 625 (8, 20, 12 and 4 of them modulo $N\mathbb{Z}[i]$), at which $\min_\gamma \|\operatorname{Re}(\bar{c}_k\gamma)\| = 17/56$. But the constant $56/17$ comes from the first levels only: by the same computations, for $\theta > 333/1106$ the set $S_{5^5}^\theta$ has only the components of Proposition 7, and the threshold of the induction step is $3303/10981 < 333/1106$ for $4 \leq k \leq 6$, hence at most that for $k \geq 6$; so the levels $k \leq 5$ and the threshold at $k = 6$ already give $\chi_c(F^2) \geq 1106/333 \approx 3.3213$.

(2) The method cannot reach beyond $10/3$. For every k the point $c_k = Nh + \frac{\rho^k}{5} - 5^{k-1}(2-i)$ lies in $S_N^{3/10}$, with $\min_\gamma \|\operatorname{Re}(\bar{c}_k\gamma)\| = \frac{3}{10}$: for $-k \leq j < k$, $5^{k-1}(2+i)\rho^j \in \mathbb{Z}[i]$ and $\bar{c}_k\rho^j \equiv h + \rho^{j-k}/5 \pmod{\mathbb{Z}[i]}$, at distance $\frac{1}{5}$ from h ; and $\bar{c}_k\rho^k \equiv h + \frac{1}{5} - \frac{2+i}{5} = h - \frac{1+i}{5}$, because $(2+i)^{2k+1} \equiv 2+i \pmod{5}$. But its distance to NE_c is at least $\sqrt{5} \cdot 5^{k-1} - \frac{1}{5}$, and to NE_q at least $\frac{7}{6}5^{k-1} - \frac{1}{5}$. So for $\theta \leq 3/10$ the probe has characters of a third, 5-adic, kind. On the plane over a number field the places above 5, which split in $F(i)$, should exclude them, but the method would have to be extended to include them and localise at 5.

(3) The least θ above which $S_{5^k}^\theta$ has only the components of Proposition 7 is $17/56$ for $k \leq 4$, $333/1106$ for $k = 5$, $3303/10981$ for $k = 6$ and about 0.30016093 for $k = 17$. We expect that it tends to $3/10$, as do the thresholds of the induction step (about 0.300036 for $20 \leq k \leq 22$); that would give $\chi_c(F^2) \geq 10/3$ in Theorem D.

10. COMPUTER CHECKS

The programs are in the folders `data/number_fields/three_colours/` and `data/number_fields/circular/` of [9], with tests in `tests/test_three_colours.py`.

Lemma 2 as a test. For a finite V , the possible values $z = 6\varphi(v)$ form the lattice $\mathcal{M} = (\text{column space}) \cap \mathbb{Z}^{2m}$, and the types are conditions on z modulo 2 and 3; a program enumerates $\mathcal{M}/2\mathcal{M} \times \mathcal{M}/3\mathcal{M}$ and decides whether some β exists. A second implementation in PARI/GP [20] uses another lattice routine, and a referee wrote a third, which agreed on all 78 of its trials. The real quadratic cases of [11] come out infeasible; $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{7})$ stay feasible; $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ is infeasible already with five vectors, $1, (7 + 4\sqrt{2}i)/9, (23 + 10\sqrt{2}i)/27, (1 + 3\sqrt{7}i)/8$ and $(9 + 5\sqrt{7}i)/16$; the sextic field $\mathbb{Q}(2\cos\frac{2\pi}{7}, \sqrt{7})$ is infeasible with 44 vectors, the obstruction being that of Lemma 7 at its two places above 3 of residue degree 3; $\mathbb{Q}(\sqrt{3}, \sqrt{7})$ stays feasible. A referee tested 23 random quartic and sextic fields: every field predicted to be 2- or 3-colourable stayed feasible, and infeasibility was found for all three quartic fields and four of the six sextic fields predicted to need four colours (for the other two the random search found no obstruction; the test is one-sided). The obstructions appear only when V contains vectors that are large at the places above 2 and 3 that split in $F(i)$, as the proof of Lemma 4 suggests.

Exact certificates. The 50 vectors $G_{25}\{1, u_2, \bar{u}_2, u_7, \bar{u}_7\}$ of Section 7 (one per pair $\pm u$), and the 70 vectors $G_{125}V$ for the five vectors above, admit no character into $[1/3, 2/3]$: branch-and-bound trees over the integer values of the relations, with exact Farkas vectors at the leaves (1579 and 11357 nodes), accepted by a separate exact checker, which also verifies that every vector has length 1 and rejects corrupted copies. With Theorem W^+ this proves $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{7})^2) \geq 4$ without Proposition 1 or Sections 3–5; a place above 7 with residue field $\mathbb{F}_7$ gives $\chi \leq 4$. The certificates of Theorem C are of the same kind, for the open interval $(2/7, 5/7)$: a branch covers all integers strictly inside the range of its relation (relations must be nonzero), and a leaf is closed by a Farkas vector valid on the open box. They have 12850 leaves ($d = 11$) and 10241 leaves ($d = 35$), and two exact checkers that share no code accept them; the second one also rebuilds $U = G_{25}V$ from d . Both reject corrupted copies. The certificate of Proposition 4 ($d = 59$, the interval $(15/53, 38/53)$) has 127181 nodes and 105761 leaves; both checkers accept it.

The lemmas. Lemma 7 was checked exhaustively for $f = 1, 3, 5$, and by the referees with their own programs (for $f = 3$ over all 3^{12} $\mathbb{F}_3$ -linear maps); the uniqueness step for odd $f \leq 79$; Lemma 6 on the finite models $\mathbb{F}_{q^2}[\pi]/(\pi^e)$ ($f = 1, e \leq 7$, and $f = 3, e \leq 2$); Lemma 5 over all μ_4 -stable subsets of $\mathbb{F}_9$. For the criterion, a PARI/GP script computes (a) and (b) from a defining polynomial; for squarefree $|d| < 200$ it reproduces the rule for quadratic fields, and it agrees with the known values $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$ [8, 18] and $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$ [10, Theorem 1]. For Section 8, two programs written separately compute κ_1 for every (p, f) quoted there, and the bound $|S(b)| \leq 2\sqrt{q}$ and the identity of $S(b)$ with a Kloosterman sum were checked numerically for small (p, f) ; Proposition 5 was checked on the finite models $O_w/\mathfrak{m}^k$ for $\mathbb{Q}_3, \mathbb{Q}_7, \mathbb{Q}_{11}, \mathbb{Q}_3(\sqrt{3}), \mathbb{Q}_7(\sqrt{7})$ and $\mathbb{Q}_{27}$.

Theorem D. The programs are in `data/number_fields/circular/probe/`; two referees wrote their own implementations, which share no code with them or with each other, in its subfolders `indep/` and `indep2/`. All use exact rational arithmetic and compute S_N^θ level by level as a union of convex rational polygons: each piece of level $k - 1$, translated by the 25 elements of $\frac{N}{5}\{0, \dots, 4\}^2$, is cut along the strips where the coordinates of $\bar{c}\rho^{\pm k}$ lie in $[\theta, 1 - \theta] + \mathbb{Z}$; a direct cell-by-cell computation agrees with this lifting for $k \leq 3$. Linear-programming values (the thresholds of the induction step, the least s in the base case, the maximum of $\min_\gamma \|\operatorname{Re}(\bar{c}\gamma)\|$ over a component) are floating-point optima certified by exact primal and dual solutions. The implementations agree on everything quoted in Section 9: the base case (least s equal to 0, 11/56 for the eight classes with exactly one integral $\nu_\pm$, and 1/4, 2/7, 3/8, 1/6 for the four orbits with the shifts of the proof; with all shifts the third value is 1/4), the polygons P and $X(\varepsilon)$, the sets S_N^θ for small k at several values of θ in $(17/56, 1/3]$, the extra points at $\theta = 17/56$, the thresholds of the induction step, and the computations at 333/1106 and at θ^* (the same number of components at every level $k \leq 19$). One referee also tested membership in S_N^θ exactly at 350000 random points ($N = 5, 25$), finding none outside the components of Proposition 7; the 5-adic points c_k of the Remark were checked exactly for $k \leq 15$.

11. QUESTIONS

- (1) *Circular local-global.* Is $\chi_c(F^2) \in \{2, 3, 7/2\}$ whenever $\chi_c(F^2) < 4$? More precisely: is $\chi_c(F^2) = 7/2$ exactly when (a) and (b) fail and some place of F above 7 has residue degree 1, and $\chi_c(F^2) \geq 4$ when (a) and (b) fail and no such place exists? By Corollary 5 this is what a local-global principle for circular colourings would say; Theorem B with Corollary 1 is its part at 2 and 3. For $\mathbb{Q}(\sqrt{23})$, where 7 splits, so that $\chi_c \leq 7/2$ by Proposition 3, it predicts $\chi_c = 7/2$; for $\mathbb{Q}(\sqrt{59})$, where 7 is inert and $\chi = 4$ [11, Corollary 2], it predicts $\chi_c = 4$. Theorem D is its part below 56/17 (below 3.3315 with the computer-assisted step). The method of Section 9 cannot go beyond 10/3, because of the 5-adic characters of the probe (Remark in Section 9); a proof along these lines of the gap up to 7/2 would have to include them, for instance by localising at the places above 5, which split in $F(i)$.
- (2) *Four colours.* Is $\chi(F^2)$, in general, the least number of colours of a locally constant colouring of one completion? (For real fields this is [13, Question 8], which also allows

Borel colourings at the real places.) Theorem B answers this at three colours. The first open case is $\mathbb{Q}(\sqrt{47})$, where $4 \leq \chi \leq 5$ ([11, Corollary 2] and [19]).

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